Matrix inequalities and norms

Bourin's crossed-Heinz inequality

Difficulty: extreme **Importance:** interesting to the community

Problem statement

For every integer $n \geq 1$, positive definite $A, B \in \mathbb{C}^{n\times n}$, real $t \in [0,1]$, and unitarily invariant norm $|||\cdot|||$, is

$$|||A^tB^{1-t} + B^tA^{1-t}||| \leq |||A+B|||?$$

Powers use the spectral functional calculus. A norm is unitarily invariant when $|||UXV||| = |||X|||$ for all unitary U, V. The positive definite formulation avoids ambiguity at exponent zero; for $0 < t < 1$ it includes the positive semidefinite case by continuity.

Why it matters

This asks for a sharp bound on noncommuting products of fractional matrix powers. Bounds of this kind support analysis of positive definite matrix computations and the comparison of matrix means.

References

1. J.-C. Bourin, *Matrix subadditivity inequalities and block-matrices*, International Journal of Mathematics 20(6) (2009), 679–691; the crossed-Heinz question. Preprint, DOI.
2. F. Kittaneh and É. Ricard, *On a question of Bourin*, Linear Algebra and its Applications 710 (2025), 356–362; central parameter interval. DOI.
3. T. Zhang, *Bourin-type inequalities for τ-measurable operators in fully symmetric spaces*. arXiv:2602.00358v1 (30 January 2026), §1, equation (1.3), Theorem 1.3. Primary text.

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The bound with constant one is proved for $t \in [1/4, 3/4]$; the endpoints $0, 1$ are immediate for positive definite matrices. The displayed assertion remains unresolved in the two outer open intervals. The latest 2026 source gives a larger explicit constant there. Searches included *Bourin crossed Heinz conjecture 2026*, *On a question of Bourin Kittaneh Ricard*, and *Bourin inequality proof counterexample 2025 2026*. No resolution of the remaining norm inequality was found. The stronger assertion for individual singular values has counterexamples and is not the present problem.